

Designing Markets

Chapter 4: Handbook of Market Design

Summary Presentation

February 2, 2026

Overview: Three Goals of Market Design

- 1 **Diagnose** potential market failures and identify their causes
- 2 **Evaluate** markets and compare alternative designs (qualitative and quantitative effects)
- 3 **Propose** new, improved designs building on lessons learned

Key insight: Seemingly small details about market rules can have large impacts on market performance

4.1 Diagnosing Market Failures

Common types of failures documented:

- **Market Power:** Exercise by large participants (US Spectrum: 7-20% higher payouts)
- **Collusion:** Centralized mechanisms viewed with suspicion (medical residency lawsuit)
- **Rent Seeking:** Wasteful competition for rents (HFT, ticket brokers)
- **Participation/Entry:** Adding one bidder $>$ optimal design (Bulow-Klemperer)
- **Fragmentation:** Kidney exchange fragmentation reduces transplants 30%+
- **Search Costs:** Students applying to too few/many schools
- **Strategic Misreporting:** Manipulable mechanisms $\rightarrow$ inefficiency

4.2 Evaluating and Comparing Designs

Key Applications:

School Choice:

- NYC reform: centralization achieved 80% of potential gains
- Deferred Acceptance vs. Immediate Acceptance comparisons
- Primary gains from coordination, not specific mechanism choice

Dynamic Assignment:

- Organs, public housing, hunting licenses arrive over time
- First-come first-served vs. last-come first-served depends on preference structure
- Use administrative data to estimate preferences and design priorities

4.2 Non-Utilitarian Objectives & Policy Evaluation

Beyond welfare maximization:

- Public housing: targeting neediest vs. allocative efficiency
- Organ allocation: survival outcomes vs. urgency
- Distributional concerns across demographics
- Meritocratic admissions and long-term outcomes

Key Challenge: Estimating treatment effects when choices correlate with outcomes

Policy Evaluation: Centralized marketplaces provide ideal data for studying broader questions in health and education economics

4.3 Proposing New Designs: Medical Match

National Residency Matching Program (NRMP)

- Matches 25,000 medical graduates to residency positions annually
- Remarkably, NRMP independently discovered Gale-Shapley algorithm in 1950s!
- Before centralization: market unraveled with offers years in advance
- Roth-Peranson (1999): Modified algorithm to handle married couples

Note: Direct welfare evidence elusive, but algorithm maintained for decades

4.3 Proposing New Designs: School Choice

Key contributions:

- Abdulkadiroğlu-Sönmez (2003): Introduced school choice as market design problem
- Documented Boston mechanism's strategic manipulability
- Proposed Deferred Acceptance and Top Trading Cycles as alternatives

Design takeaways:

- Equilibrium efficiency differences between well-designed mechanisms are small
- Complexity of strategic behavior and fairness costs matter more
- Deferred Acceptance widely adopted; Top Trading Cycles less so (priority interpretation issues)

4.3 Proposing New Designs: Course Allocation

A-CEEI Mechanism (Budish):

- Problem: Allocate oversubscribed courses efficiently and fairly
- Draft mechanisms common but strategically manipulable
- A-CEEI: Approximately ex-post efficient, approximately strategy-proof
- Satisfies envy-freeness and maximin share fairness

Multi-paper research program:

- 1 Theory + data on HBS draft mechanism
- 2 Propose A-CEEI with computational algorithm
- 3 Laboratory experiment at Wharton
- 4 Implementation in practice

4.3 Proposing New Designs: Food Banks & Financial Exchanges

Feeding America Pseudomarket (Prendergast):

- Internal currency allocated proportionally to need
- Food banks bid for donations; winners pay, losers receive currency
- Results: More equitable, efficient sorting, vertical differentiation

Frequent Batch Auctions (Budish et al.):

- Replace continuous limit order books in financial markets
- Eliminate speed-based arbitrage rents and arms race
- Could reduce liquidity costs by 17% (\$5B/year in equities)
- Trades on price rather than microsecond speed advantages

Key Takeaways

- 1 **Details Matter:** Small rule changes can have large impacts
- 2 **Coordination $>$ Optimization:** Moving from uncoordinated to sensible design matters more than optimal mechanism choice
- 3 **Multiple Objectives:** Real-world design must balance efficiency, fairness, simplicity, and participation
- 4 **Empirics Essential:** Structural models and experiments crucial for comparing mechanisms and quantifying effects
- 5 **Iterative Process:** Successful designs often require multiple papers using different methodologies